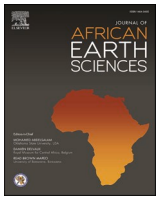
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Paleomagnetic evidence of early Pleistocene counterclockwise rotation in the Butajira volcanic zone, central Main Ethiopian rift

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ABSTRACT

A total of 208 paleomagnetic core samples were collected from twenty-seven sites within the Butajira volcano chains of the central Main Ethiopian rift (MER), providing insights into tectonic rotation. Core samples were taken from ignimbrite (~1.7–2.54Ma), basalt, and trachyte rock units (<1.6Ma) along the western margin of the central MER. Each core sample underwent analysis using twin specimens subjected to stepwise alternate field (AF) and thermal (Th) demagnetization techniques. Furthermore, a single specimen from every site was employed for rock-magnetic experiments. Detailed rock magnetic experiments identified titano-magnetite, magnetite, and hematite as the remanence carrier magnetic minerals. The natural remanent magnetization (NRM) directional analysis revealed two components:1) a low-stability (LS) component removed by low-temperature (<300 °C) and low-field (<20 mT) and 2) a high-stability (HS) with heating >300 °C or 20 mT increments showed vectors pointing to the plot's origin, indicative of the characteristic remanent magnetization (ChRM). All sites, except one, displayed reverse polarity NRM. The tilt-corrected mean direction calculated from 19 sites, yielded $D_o = 355.7°$, $I_o = 9.9°$ ($N = 19$, $K = 22.9$, $\alpha_{95} = 7.3°$), indicates a $6.2° \pm 6.2°$ counterclockwise (CCW) tectonic rotation relative to the reference dipole geomagnetic field direction of 2Ma $D_x = 1.9°$, $I_x = 15.5°$ ($N = 32$, $K = 105.6$, $\alpha_{95} = 2.5°$) for stable Africa (Besse and Courtillot, 2003). The oblique rifting and pure-dip slip kinematics that characterize the border faults in the study area could have caused this minor counter-clockwise rotation and agrees with analogue model prediction of Corti et al.,2013.

1. Introduction

The East African Rift System (EARS) is a geologically significant rift system consisting of distinct rift sectors with and without surface magmatism, extending from the Indian Ocean and southern Africa to the Arabian Peninsula along the eastern margin of Africa. The EARS has advanced to seafloor spreading in the Red Sea (RS) and Gulf of Aden (GA) rift in northern Ethiopia and the Arabian Peninsula while also experiencing continental rifting in the northern Main Ethiopian Rift (MER) (e.g., Courtillot et al., 1999; Wright et al., 2006) (Fig. 1a–and b). The MER is a significant part of the EARS, connecting the Afar Depression at the junction of the RS and the GA with the Turkana Depression and Kenya Rift to the south (e.g., Mohr, 1983; Rosendahl, 1987; Braile et al., 1995; Boccaletti & Peccerillo,1999; Chorowicz, 2005).

The MER is known for its volcanic activity and serves as a record of the phases of rift development, from its inception to the separation of continental masses and the creation of early oceanic spreading (e.g., Ebinger, 2005). The MER is a prime location to investigate continental extension, plate rupture, rotational deformation, and the concentration of continental deformation at the oceanic spreading center (Corti, 2009; Kidane et al., 2009). The southern, central, and northern sections of the MER exhibit intricate tectonic and volcanic processes, highlighting its complexity (e.g., Boccaletti et al., 1999; Ebinger and Casey, 2001; Casey et al., 2006; Abebe et al., 2007; Kurz et al., 2007; Beutel et al., 2010; Rooney, 2010; Rooney et al., 2011; Keir et al., 2015; Hutchison et al., 2016). This study paleomagnetic study focuses on the Butajira Volcanic

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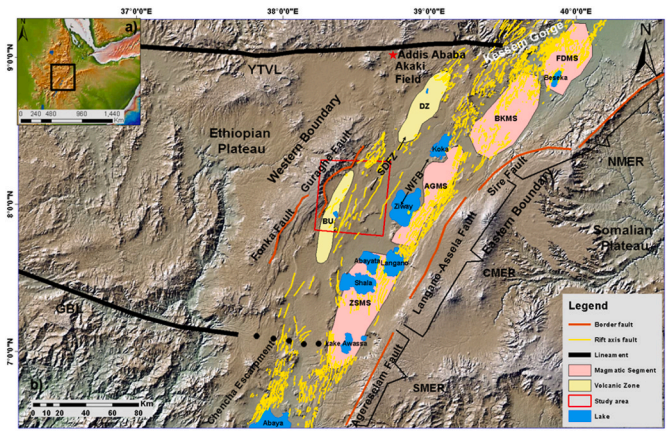


Fig. 1. a) Digital elevation model (DEM) showing the study of CMER; b) Blown-up DEM illustrating the tectonic and structural characteristics of the MER sectors modified after Agostini et al. (2011b) and Corti et al. (2018). Major faults along the rift margins and rift axis are represented by bold lines and regular lines, respectively. Active volcanic centers of each magmatic segment are highlighted in pink, and lakes are depicted in blue. The study area is highlighted by a solid red rectangle. NMER: Northern MER; CMER: Central MER; SMER: Southern MER; YTVL: Yerer Tulu Welel Volcanic lineament; GBL: Goba– Bonga lineament; SDFZ: Silti Debre Zeyt Fault zone; Bu: Butajira Volcanic Zone; DZ: Debre Zeyt Volcanic Zone; WFB: Wonji Fault Belt; ZSMS: Ziway-Shala magmatic segment; AGMS: Aluto-Gedemsa magmatic segment; BKMS: Boseti-Kone magmatic segment; FDMS: Fantale-Dofan magmatic segment.

zone within the Central MER, characterized by a less advanced stage of rifting, to evaluate the tectonic rotation of crustal blocks (Figs. 1 and 2). Here, significant deformation is localized at large boundary faults, with axial deformation in an early stage only (Agostini et al., 2011a, 2011b). In this sector of the MER, volcanism and magmatic modification of the crust/lithosphere are less developed (Keranen and Klemperer, 2008). Geological, geophysical, and structural data suggested a complex history

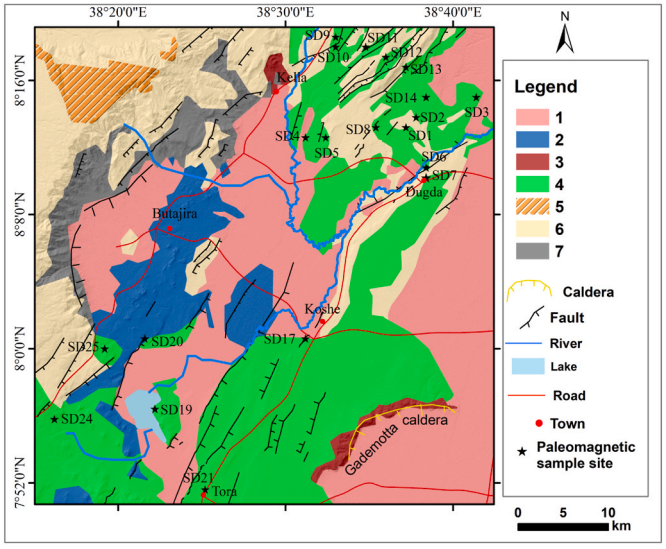


Fig. 2. Simplified geological map of the Butajira area, modified after Abebe et al. (2010) and (Molin&Corti,2015) (1) Lacustrine sediments (2) Wonji Basalts (<1.6 Ma), (3) Quaternary lava flow, Bora-Berich rhyolites (<1.6 Ma), (4) Pyroclastic deposits including ignimbrite (2.54–1.7 Ma),(5) Late Miocene-Pliocene GashMegal rhyolite,(6) Miocene–Pliocene rift-shoulder trachytic–rhyolitic volcanics and pyroclastic deposits (5.2–2.6 Ma), (7) Pre rift rocks: Precambrian metamorphic rocks, Mesozoic cover, (Oligocene-Early Miocene basaltic rock: 32–29 Ma), (Guraghe Basalts: 10.6–8.3 Ma).

of rifting of MER potentially linked to a change in Nubia-Somalia motion that occurred ~6.6 to 3Ma (Wolfenden et al., 2004) or at the Pliocene–Quaternary boundary (Bonini et al., 1997, 2005; Boccaletti et al., 1998, 1999). Plate kinematics models support these findings, suggesting a shift in the pole of rotation between Nubia and Somalia (i.e., a change in kinematics in the EARS during the Pliocene (Lemaux et al., 2002). Due to the regional clockwise rotation of the Somalian plate (e.g., Saria et al., 2013, 2014; Stamps et al., 2021) and oblique rift kinematics of MER (Corti, 2009; Corti et al., 2013), the Central MER exhibits distinctive features that make it an excellent location for studying extensional processes, fault geometries, and kinematics.

Paleomagnetic studies (Muhabaw et al., 2022; Kidane et al., 2006 and Nugsse et., 2018) indicate varying degrees and directions of crustal tectonic rotation, suggesting that oblique rifting, as depicted in Fig. 1, has induced vertical-axis rotation of crustal blocks along with associated strike-slip deformation (Brune et al., 2018; Zwaan and Schreurs, 2020). Different kinematic models, along with paleomagnetic and geological data, offer conflicting tectonic evolution of the MER.

In his early work, Di Paola (1972) argued for an E-W extension of the rift during its entire history, a notion supported by WoldeGabriel et al., 1990. A follow up study using evidence from remote sensing and geographic information systems (e.g., Chorowicz et al., 1994; Korme et al., 1997) proposed a NW-SE extension of the rift. Bonini et al. (1997) based on laboratory experiments suggested a change in extension direction by 45° occurred during the Quaternary from being orthogonal to an oblique to the rift shoulder; this interpretation is in par with field based structural observations of a change in extension direction from NW-SE to E-W during the Quaternary (Abebe, 1993; Boccaletti et al., 1994), both studies imply a block rotation about a vertical axis. Keranen and Klemperer (2008) interpret the right stepping en-echelon rift-axes (magmatic segments) as a nascent mid-oceanic ridge where blocks in the overlap zone of the rift-axes are bound to rotate about a vertical axis. Chernet et al. (1998) suggested existence of oblique slip transform faults or accommodation zones that are oriented transverse linking offset rift axes segments, like transform faults in mid-oceanic ridges. Recently, detailed field structural investigation revealed that the boundary faults always show a dip slip kinematics with only local slight component of strike slip <20° Agostini et al., 2011a and b; Corti et al., 2020). The analogue models by Corti et al., (2013), show very well that this is due to a reorientation of the stress along the faults. This reorientation implies slight counterclockwise rotation for the entire rift valley length.

Paleomagnetic studies on fault-bounded blocks in the Fentale and Dofan magmatic segments (Kidane et al., 2009; Nugsse et al., 2018) revealed ~ 7° counterclockwise block rotation associated with transtensional deformation, aligning with the localized deformation observed in these magmatic segments (Casey et al., 2006). Corti et al. (2013) suggest an analogue model indicating an inevitable minor counterclockwise rotation, linked with oblique rifting, a model supported by paleomagnetic experiments and analyses of extensional faulting along oblique rift margins. On the other hand, Muhabaw et al. (2022) report a recent paleomagnetic study of Gedemsa MS in Central MER, indicating a 17° clockwise block rotation, which he modeled and attributed to the opening of DebreZeit-Butajira (DzB) and Yerer – Tullu Wellel Volcano-tectonic Lineament (YTVL) segments (Fig. 1b) (see Muhabaw et al., 2022 for details), which reiterated even though a rift wide slight counterclockwise rotation is bound to result from reorientation of the stress along the faults (Corti et al., 2013), other local and regional structural elements play a role in vertical axis block rotation. The CMER, the sector where this study is done, has rifted for ~ 6Ma (Bonini et al., 2005; Keranen and Klemperer, 2008), but a complete breakup has not happened yet, suggesting a complex geological and tectonic history in the region. This study aims to elucidate the tectonic movements linked to Pull-apart zones, en-echelon tensional fissures, and right-stepping arrangements of boundary fault segments of the central MER using paleomagnetic techniques (Fig. 1b). The study aimed to answer two main questions: 1) Does the recent lava flow in the Western CMER

exhibit any indications of crustal block rotation since their emplacement? 2) What is the magnitude and direction of crustal block rotation in the Butajira volcanic zone with respect to vertical axis rotation? 3) In what geodynamic context are the block rotation possible?

2. Geologic and tectonic setting of MER

The MER is a segment of the EARS, which consists of 80–100 km-wide rift valleys stretching thousands of kilometers across Eastern Africa (Fig. 1). The MER forms the incipient plate boundary between Nubia and Somalia, whose relative motion is ~95°N–100° E at rates of ~4–6 mm/yr (e.g., Fernandes et al., 2004; Stamps et al., 2008; Kogan et al., 2012; Knappe et al., 2020; Saria et al., 2014) in a roughly E-W direction (Fig. 1). MER is composed of three main segments (Northern MER, Central MER, and Southern MER) differing in terms of rift trend, fault timing and patterns, and lithospheric characteristics (Fig. 1) (e.g., Mohr, 1983; Hayward and Ebinger, 1996; Bonini et al., 2005; Corti, 2009). East-west trending prominent volcano tectonic lineament of YTVL that affects the western escarpment intersect the Northern MER east of Addis Ababa in the Kessem Gorge (Fig. 1b). The Goba-Bonga volcano tectonic lineament or GBVL is another East-west oriented major structure cutting through the western escarpment as well and it extended in to the rift valley where its expression is represented by the East-West alignment of the Hawassa and Corbetti calderas (Corti et al., 2018) dividing the CMER from the SMER (Fig. 1b), characterized by alignments of normal faults and major volcanoes (Abebe et al., 1998; Abbate and Sagri, 1980; Corti et al., 2018). Although, Goba is on the eastern escarpment side, no straightforward evidence of the lineament is described (Corti et al., 2018).

As far as faulting is concerned, the faulting pattern in the MER is characterized by two distinct systems of normal faults (Fig. 1): 1) Border faults and 2) a set of faults affecting the rift floor, often referred to as the Wonji Fault Belt (WFB), (Boccaletti et al., 1998; Mohr, 1962; Gibson, 1969; Corti, 2008; Agostini et al., 2011b; Corti et al., 2013). The fault systems across different sectors of the MER (Fig. 1) vary in orientation, structure, timing of activation, and their connection to magmatism (Hayward and Ebinger, 1996). The WFB is characterized by a series of closely spaced, short, active axial faults with associated extension fractures and grabens, which display minimal vertical displacement (Boccaletti et al., 1998; e.g., Acocella et al., 2003; Agostini et al., 2011a). The WFB, developed in the last 2 Ma, comprises sigmoidal overlapping right-stepping fault segments, influencing the rift obliquely: oriented ~ N20° E in the NMER and N12° E in the CMER (Corti, 2009). The WFB are prominent in the northern MER but diminish in occurrence towards the south. Border faults commonly exhibit considerable length, wide spacing, and significant vertical displacements (e.g., Boccaletti et al., 1998; Agostini et al., 2011a). Border faults in the MER are prominent from NMER to SMER, mostly inactive and eroded in NMER (Wolfenden et al., 2004). The boundary faults developed between 11 and 6 Ma, stayed active until about 2Ma, and are now primarily inactive in the NMER sector (Corti, 2009; Agostini et al., 2011a and b; Corti et al., 2020). The rift matured northward, with NMER's border faults largely inactive seismically, while CMER and SMER's faults remain active (Casey et al., 2006; Keir et al., 2006; Wolfenden et al., 2004; Agostini et al., 2011a and b; Molin and Corti, 2015). Over the past 2 Ma, deformation in the CMER to NMER shifted from NE-trending border faults to the rift floor, concentrating on NNE-striking magmatic segments arranged in a right-stepping en echelon pattern (Ebinger and Casey, 2001; Casey et al., 2006). In the CMER, the rift valley trends between N25°E and N45°E is characterized by major rift escarpments on both western and eastern margins (Figs. 1 and 2). The Guraghe and Fonko faults mark the western margin (current study area), trending N25°E to N35°E and dipping to the ESE. In contrast, the eastern margin is defined by the Asela–Langano fault system, trending N30°E and dipping to the WNW (Corti, 2009; Corti et al., 2018; Corti et al., 2020).

In the MER, Quaternary-Recent volcanism is primarily characterized by rhyolites, ignimbrites, pyroclastic deposits, and to a lesser extent, basalts (WoldeGabriel et al., 1990; Gasparon et al., 1993; Boccaletti et al., 1998; Trua et al., 1999; Peccerillo et al., 2003; Rooney et al., 2012; Giordano et al., 2014). Magmatic activity is concentrated along the WFB and the rift marginal Silti-Debre Zeit Fault Zone (SDFZ) belt, with mafic volcanism occurring along these zones in the form of numerous monogenetic basaltic vents, such as spatter cones, scoria cones, and maars (Rooney et al., 2007, 2014a; Maccaferri et al., 2014; Mazzarini et al., 2013). The central MER (Fig. 1) exhibits reduced tectonomagmatic activity proceeding southward from the northern MER sector (Keranen and Klemperer, 2008). The CMER (Fig. 1) is the primary area of interest in the current paleomagnetic study, with concentrated tectonic-magmatic activity occurring along two subparallel belts: the WFB in the east and a volcanic belt extending from Silti-Butajira to Debre Zeyt in the western. The WFB has intense faulting, extension, and a robust magma plumbing system, while the SDFZ belt has less developed magmatic activity and lacks significant surface faulting (WoldeGabriel et al., 1990; Rooney et al., 2007, 2011). The central rift floor shows no notable faulting or volcanic activity, indicating less significant axial tectonic and magmatic deformation in this region (Agostini et al., 2011b).

The geology of Butajira area consists of a wide range of lithologies exposed along the rift margin, stream valleys, and road cuts: pre-rift; syn-rift and post-rift sequences. Pre-rift sequences outcrop along the Guraghe border faults, which include the oldest Precambrian crystalline basement rocks (biotite–alkali feldspar gneisses), which are cut by quartz–feldspathic pegmatite veins and the Mesozoic clastic sediment cover (Fig. 2 – unit7) that unconformably overlies the basement and outcrop at Kella faulted block (WoldeGabriel et al., 1990; Abebe et al., 2010). This sequence is unconformably overlain by Oligocene-Early Miocene olivine-tholeiites Kella basalts dated at 32–29 Ma (Fig. 2, unit7) (e.g., Bonini et al., 2005), and are believed to be associated with the Oligocene-lower Miocene basaltic sequences of the Trap Series (e.g., Mohr and Zanettin, 1988; Hoffman et al., 1997). The Guraghe-Anchar basalts (10-8.3Ma) non-conformably overlie the pre-rift sequence and outcrop along the Guraghe margin (Fig. 2, unit7), Abebe et al. (2010). Syn-rift units include pyroclastic rocks accompanied by trachytic and rhyolitic lava domes and flows with an age range of 5–2.6 Ma cover the MER shoulders and rift floor (Fig. 2, unit6). The Late Miocene-Pliocene Gash Megal rhyolite and are exposed along the rift shoulder of the north Guraghe border fault (Fig. 2, unit5) (e.g., Abebe et al., 2010; WoldeGabriel et al., 1990). The post-rift sequences are volcanics and sediments that fill the floor of the rift and chiefly are pyroclastic deposits including ignimbrite aged 2.54–1.7 Ma are well exposed along the marginal graben and rift floor (Fig. 2, unit 4). Along the rift floor, Quaternary lava flow of Bora-Berich rhyolites (Fig. 2, unit3) aged <1.6 Ma (Bonini et al., 2005; Abebe et al., 2010), Wonji volcanics <1.6 Ma, that include pumice fall and flow deposits, rhyolitic–trachytic lava flows forming central volcanic edifices, fissural basaltic lava flows with related to scoria and phreatomagmatic cones (Fig. 2, unit2) are exposed, and relatively thin veneer of lacustrine deposits (Fig. 2, unit 1) covered much of the top of rift floor of Western CMER.

3. Paleomagnetism

3.1. Sampling and laboratory methods

Paleomagnetic sampling targeted the western margin of the CMER, known for the N25°E–N35°E-trending and ESE-dipping Guraghe and Fonko faults. Core samples were collected from ignimbrites (1.7–2.54Ma) and trachytes <1.6Ma (Abebe et al., 2007), exposed along fault scarps, stream gullies, and road cuts (Fig. 2). In total, 208 core samples were collected from twenty-seven sites (Fig. 2), comprising ignimbrite (21 sites), basalt (5 sites), and trachyte (1 site) from the western marginal Graben and Butajira volcano chains in CMER (Fig. 2). The samples were drilled using a portable gasoline-powered drill cooled

by water and oriented in situ using magnetic compass and when the area is not under a shadow, we also used Sun compass. Each site yielded 6–8 core samples. The oriented core samples, with a diameter of 2.5 cm, were subsequently sliced into standard 2.2 cm long twin specimens using a dual-blade core slicing saw for further analysis.

Three hundred eighty-three specimens (383) underwent stepwise demagnetization using thermal (Th) or alternating field (AF) methods. Thermal demagnetization reached 680 °C in twenty steps, with 20–50 °C increments applied to six specimens per site. Alternatively, AF demagnetization involved several narrow steps, up to a maximum of 120 mT, in ten steps for the remaining eight specimens per site. Both methods aimed to isolate the characteristic remanent magnetization (ChRM) by eliminating one or two soft secondary magnetization components. However, AF demagnetization proved more effective in isolating the ChRM directions of volcanic products in the Central MER. The Paleomag6.5 software (Cogné, 2003) facilitated various analyses, including examining progressive demagnetization data, principal component analysis (PCA), structural corrections, and computation of site and formation mean to paleomagnetic data.

Individual directional components were determined via PCA of linear segments in Zijderveld plots (Zijderveld, 1967) or analysis of great circles remagnetization (Halls, 1978; McFadden and McElhinny, 1988). Fisher statistics computed the paleomagnetic site and overall mean directions (Fisher, 1953). Most of ChRM component was calculated for samples that displayed nearly linear demagnetization trajectories toward the origin of a vector plot (Fig. 5a). Directional components of remanent magnetization were grouped and averaged based on the blocking temperature and coercivity ranges in which they were isolated. Lastly, rotation and flattening values were compared to the observed paleomagnetic direction with respect to Africa, assessed according to Demarest (1983), utilizing the African stable apparent polar wander path (AAPWP) by Besse and Courtillot (2003) for two million years ago sites.

Additionally, representative specimens were chosen from each sampling site for further magnetic mineralogy analyses. The magnetic properties were investigated by analyzing the iron oxides carrying the Natural Remanent Magnetization (NRM) through IRM acquisition experiments, using an ASC-IM Impulse magnetizer with an amplitude of up to 1.2T. The acquired IRMs were then thermally demagnetized in steps, and the resulting unblocking temperature spectrum offered valuable insights into the magnetic properties of the specimens. All paleomagnetic routines, sample trimming, and directional measurements and rock magnetic experiments were conducted at the paleomagnetism laboratory of Addis Ababa University in Ethiopia using LDA-3 AF demagnetizer, MMTD-80 thermal demagnetizer, JR6-A spinner magnetometer, and ASC- impulse magnetizer.

3.2. Rock magnetic result

The paleomagnetic investigation is done in the basic paleomagnetic laboratory at Addis Ababa University. Detailed rock magnetic studies such as hysteresis cycles, susceptibility vs temperatures, and grain size analyses require access to bigger paleomagnetic laboratories in elsewhere in the world. Volcanic rocks such as those studied in Afar and MER (e.g., Kidane et al., 1999; Kidane et al., 2003; Nugsse et al., 2018) have proven to have mostly straight forward behavior upon demagnetization and the normalized Th and AF NRM intensity decay curve approximates the magnetic unblocking temperatures and when combined with IRM with thermal demagnetization they could reveal the magnetic mineralogy.

3.2.1. Normalized NRM intensity decay curves

In Fig. 3, NRM intensity decay curves of seven representative specimens demagnetized by AF are depicted alongside their corresponding Th demagnetization. The curves reveal that approximately 50% of NRM is eliminated for most specimens at temperatures ranging from 350 °C to

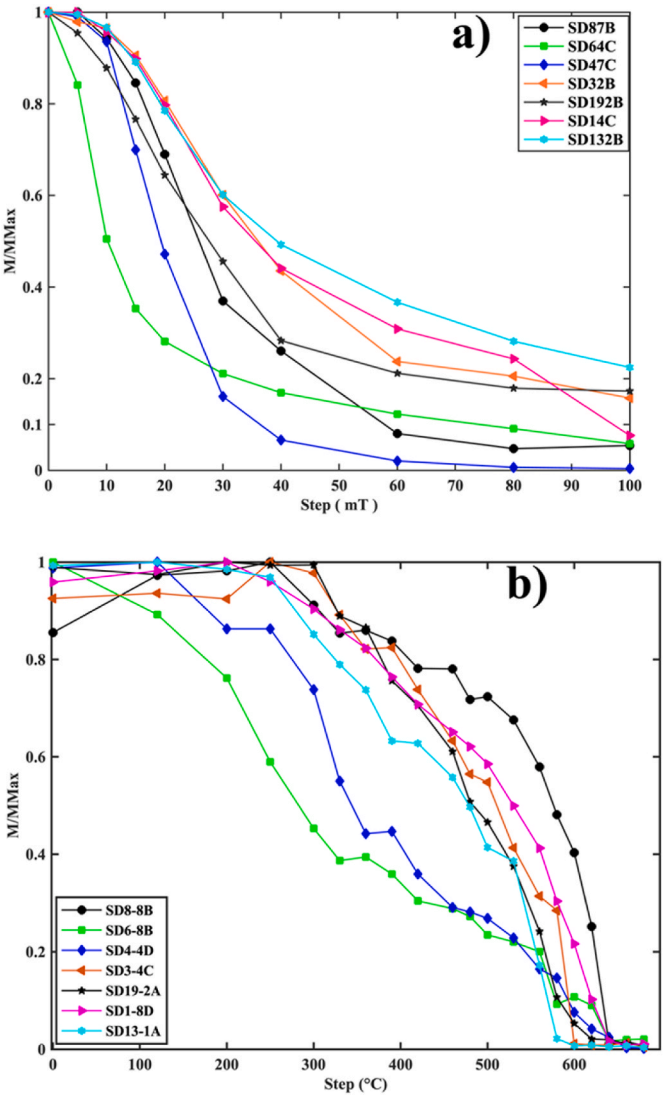


Fig. 3. Representative normalized Natural Remnant (NRM), Intensity decay curves for specimens demagnetized by demagnetization (a) Alternative field (AF) and Thermal (TH) techniques decay curve.

520 °C. However, certain specimens, like SD8-8B, could withstand temperatures up to 620 °C (Fig. 3b). Most specimens were fully demagnetized at temperatures between 580 °C and 640 °C (Fig. 3b). The median destructive field (MDF) for most specimens falls within the range of 20 mT–40 mT (Fig. 3a). Following the application of the maximum field, most specimens retained 5–25% of magnetization, except for SD4-7C, which was utterly demagnetized at 100 mT. A notable intensity drop was observed between demagnetization steps at 560 °C and 600 °C, indicating the prevalence of titanium-poor titanomagnetite as the primary carrier mineral. Some specimens experienced magnetization loss during the 640 °C demagnetization step, suggesting titano-hematite as the dominant magnetic mineral in those rocks (Fig. 3b). Moreover, NRM intensity of ignimbrite strongly magnetized ranges from 0.0207A/m (SD24) to 6.3A/m (SD3), indicative of varying magnetic mineralogy concentrations, which are linked to rock composition.

3.2.2. Isothermal remnant magnetization (IRM) acquisition and thermal demagnetization

The magnetic mineralogy experiment revealed diverse coercivity and Curie temperatures, suggesting multiple magnetic mineral phases, as shown in Fig. 4. Twenty-seven specimens underwent progressive

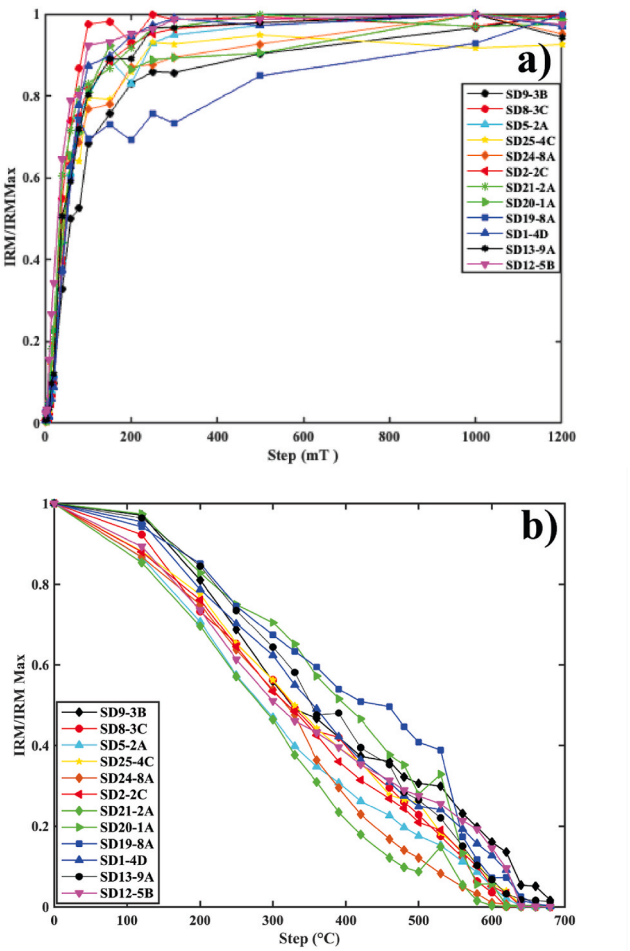


Fig. 4. Normalized Isothermal Remnant Magnetization (IRM) acquisition curve for representative sample and corresponding (a)IRM acquisition curve and (b) Thermal demagnetization.

magnetic field increases up to 1.2 T to assess their IRM acquisition patterns, followed by Th demagnetization up to 680 °C (Fig. 4a and b). Two distinct behavioral groups emerged: the first group (SD1, SD2, SD5, SD8, and SD12) exhibited rapid IRM acquisition until saturation around 300 mT, with smooth thermal demagnetization curves up to 620 °C, indicating the presence of titanium-poor titano-magnetite and magnetite as the magnetic carriers primarily. However, some specimens continued to 640 °C, suggesting a minor presence of titano-hematite. The second group (SD9, SD13, SD19, SD24, and SD25) showed gradual IRM acquisition up to 1.2T, remaining unsaturated, with thermal demagnetization curves remaining smooth until 680 °C, indicating titano-hematite as the dominant magnetization carrier. Specimens SD20 and SD21 displayed an increase in IRM with IRM demagnetization around 530 °C, possibly indicating the presence of maghemite transforming into hematite. Overall, IRM acquisition curves suggest that the dominant magnetization carriers at most sites are titano-magnetite, titano-hematite, magnetite, and hematite.

3.3. Paleomagnetic direction

Two types of demagnetization were conducted on a total of 383 paleomagnetic specimens. In Fig. 3, typical Th and AF demagnetization diagrams from reversed sites are depicted. Principal component analysis (Kirschvink, 1980) was employed to identify vector components (Fig. 5a &b). The NRM of all measured samples, treated by both AF (218 specimens, Fig. 5a left column) and Th (165 specimens, Fig. 5a right column) demagnetization methods, consisted of one or two components (Fig. 5). The initial magnetically soft component was eradicated by low-temperature (<300 °C) and low-field (<20 mT) components regarded as a secondary viscous remanent magnetization, except specimen SD12-6A required heating to 390 °C (Fig. 5a right column). For the two sets of specimens, the high temperature (>330 °C) and 20–80 mT demagnetization components, displaying nearly linear demagnetization trajectories towards the origin of a vector plot were used to isolate the ChRMs. A ChRM typically emerged between 500 °C and 620 °C indicating magnetite as the primary magnetization carrier (Fig. 5b). In the case of Th demagnetization, a ChRM is isolated between 400 and 680 °C for about 10% of the samples, suggesting the coexistence of hematite and maghemite as the main magnetization carrier (Fig. 5). The best-fit line of Principal Component Analysis (PCA) of Kirschvink (1980) was utilized if the trajectory was defined by at least four consecutive demagnetization steps (Zijderveld, 1967) with a maximum angular deviation <15° (Fig. 5a and b). However, for some samples with overlapping unblocking spectra, a ChRM vector was determined using the great circle fit method (Halls, 1976, 1978; McFadden and McElhinny, 1988) (Fig. 5c). Sister samples from each site underwent detailed AF (~218 specimens), and Th (~165 specimens) demagnetization revealed identical paleomagnetic directions (Fig. 5). The characteristic component observed in the volcanic deposits was better isolated through AF cleaning than Th demagnetization (Fig. 5a and b).

3.4. Site mean direction

Fisher statistics (Fisher, 1953) were applied to compute the site and group mean directions, utilizing the Paleomac software package (Cogné, 2003). Site mean directions derived from multiple samples were deemed acceptable for further analysis if their 95% confidence circle (α_{95}) measured less than 15° (Table 1). The NRM directional distribution shows stronger clustering within individual sites than among different sites, which resulted in a relatively large overall site mean confidence circles (α_{95}) (Fig. 6a and b). Table 1 presents the average characteristic remanent magnetization (ChRM) for one normal and eighteen reverse sites and their corresponding Fisher statistical parameters (Fisher, 1953). Site (SD4) was excluded from the mean direction analysis due to significant declination scatter, possibly reflecting local tectonic influences (Fig. 6), or transitional/excursion tendencies, which is confirmed using the recursive method proposed by Vandamme (1994). Remanence directions exhibited normal and reverse polarity (Fig. 6), suggesting a coherent grouping of intra-site remanence directions. The site mean direction of the normal polarity flow, D = 354.5°, I = −29.9° (n = 9, K = 221.6, α_{95} = 3.5°), is 180° apart from the mean direction for reverse polarity, D = 175.8°, I = −12° (N = 18, K = 29.5, α_{95} = 6.5°), (Fig. 6b).

Due to the few numbers of site (1) in the normal polarity reversal test could not be conducted but visually one can tell the normal polarity is antipodal to the reverse polarity. The overall mean direction of 19 individual sites (Fig. 6a), after reversing the reversed polarity and excluding SD 4 with anomalous direction, is Dx = 355.7°, Ix = 9.9° (N = 19, K = 22.9, α_{95} = 7.3°). Virtual geomagnetic pole (VGP) calculations from the reliable mean directions of nineteen sites, along with associated parameters, are provided in Table 1 and depicted in Fig. 6c. The overall mean VGPs of the sampled sites are (λs = 273.3°, ϕs = 84.6°, N = 19, K = 40, A_{95} = 5.4°). The Vandamme (1994) cut-off procedure identified a cluster of nineteen sites and helped exclude one site with anomalous direction from further interpretation.

4. Discussion

4.1. Tectonic rotation

To determine the amount and sense of block rotation, we compared the overall tilt corrected Butajira volcanic zone mean direction D =

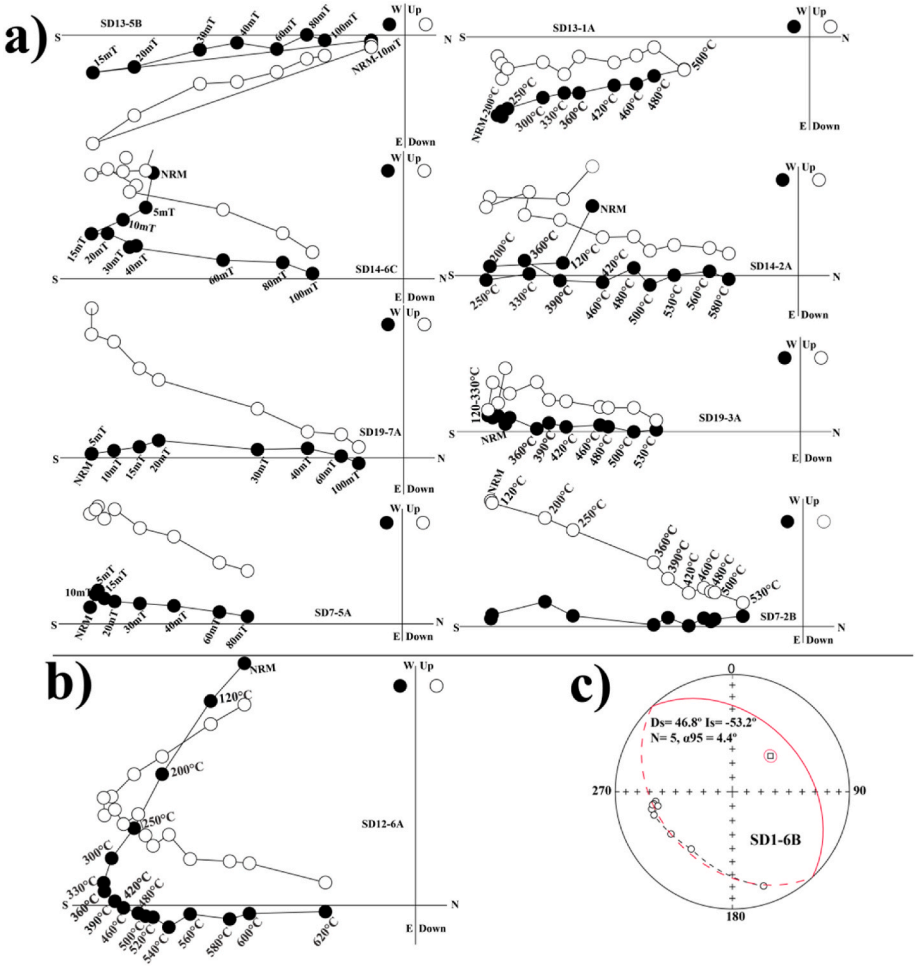


Fig. 5. Typical demagnetization behavior treated by AF (left column) and Th (right column) sister samples obtained from the same sites A mostly single NRM component of magnetization (A) and two NRM components with no overlap identified (b) remagnetization circles used to calculate ChRM directions (c). Solid and open circles dots represent projections on the horizontal and vertical planes, respectively. The numbers refer to temperatures in °C or alternating fields in mT. NRM stands for natural remanent magnetization.

355.7°, I = 9.9° (N = 19, K = 22.9, α_{95} = 7.3°) with the expected mean geomagnetic dipole reference field direction for 2 Ma, D = 1.9°, I = 15.5° (N = 32, K = 105.5, α_{95} = 2.5°), based on linear interpolation between the present day mean pole at 90° and the 10Ma pole of the African apparent polar wander path (AAPWP), by Besse and Courtillot (2003). This comparison indicates the two groupings/mean directions are statistically different both at 99% and 95% significance level (McElhinny, 1964; McFadden and McElhinny, 1990), which indicate the observed mean direction has moved with respect to the expected. Hence, the difference between the overall mean observed paleomagnetic declination of Butajira with the expected declination gives us the amount and sense of rotation, R = D_o-Dx; where R is the amount of rotation, D_o is the observed declination, D_x is the expected declination. This result indicates a counterclockwise rotation of −6.2 ± 6.2° (Fig. 6a), corresponding to a rotation rate of 3.1° ± 3.1° Myr^{-1}. We utilized Beck's method (1980) to compute the difference between the observed and reference declinations and inclinations (ΔD = −6.2 ± 6.2, ΔI = 5.6 ± 6.2). The vertical-axis rotation (R) and its confidence limits (ΔR) suggest a CCW rotation of R ± ΔR = 6.2° ± 6.2° and, a flattening (F) or inclination shallowing (I) and its confidence limit (ΔI) of I±ΔI = 5.6° ± 6.2°, where $\Delta R = 0.8(\Delta D_o^2+\Delta D_x^2)^{1/2}$ and $\Delta F = 0.8(\Delta I_o^2+\Delta I_x^2)^{1/2}$ as shown by Demarest (1983). The comparison analysis revealed that the Butajira volcanic zone rotated counterclockwise by ~6.2° relative to the stable African plate during the last 2 Ma. The inclination flattening (F), decrease of inclination from expected, although common phenomena in sedimentary rocks, which happen due to sediment compaction and diagenesis, in this study as we are dealing with volcanic rocks, the flattening is caused by the effect of long-standing non-dipole field throughout the rift at least in the Afar depression (Kidane et al., 1999, 2003) and in the MER (Kidane et al., 2009, 2010; Nugsse et al., 2018).

Fault-slip data from both rift margins in the Central MER suggest a stress field predominantly oriented ESE–WNW, with some local variations ranging between E–W and NW–SE directions (Boccaletti et al., 1992, 1998; Abebe,1993; Korme et al., 1997; Acocella and Korme, 2002; Bonini et al., 2005; Pizzi et al., 2006). Pull-apart zones, en-echelon tensional fissures, and right-stepping arrangements of boundary fault segments (Boccaletti et al., 1992, 1998; Mazzarini et al., 2013; Abebe et al., 2005; Bonini et al., 2005), suggest a sinistral component of oblique slip-on NE–SW-trending fault planes, potentially causing a current counterclockwise rotation. Moreover, Agostini et al. (2011a and b) and Corti et al. (2020) have shown that the boundary faults always show a dip slip kinematics with only local slight component of strike slip <20°. The analogue models by Corti et al. (2013), show very well that this is due to a reorientation of the stress along the faults. This reorientation implies slight counterclockwise rotation that well accord with the presented data.

Furthermore, our result is consistent with most of the prior paleomagnetic studies within the MER indicating CCW rotation in Fentale and Dofan magmatic segments (Kidane et al., 2009, 2010; Nugsse et al., 2018) including Kidane et al. (2006), when mean directions of the rocks

Table 1
Paleomagnetic site mean directions of Butajira volcanic products, expected direction for a similar age of stable Africa, and their corresponding virtual geomagnetic poles (VGPs) with statistical parameters for all analyzed rocks.

ID	Lat	Long	N	Mean directions				VGP			
				D_S	I_S	K	α_{95}	ϕ_S	λ_S	K	A95
SD-1	8.22	38.62	8	161.8	−14.1	103.1	5.5	126.5	−71.9	2.9	5.6
SD-2	8.23	38.63	6	178.4	−7.2	122.0	6.1	57.5	−85.2	3.1	6.1
SD-3	8.25	38.69	10	181.7	−15.1	398.3	2.4	328.1	−88.2	1.3	2.5
SD-5	8.21	38.54	6	190.8	−5.6	106.3	6.5	334.7	−78.0	3.3	6.6
SD-6	8.18	38.64	5	196.6	2.6	64.9	9.6	337.7	−71.0	4.8	9.6
SD-7	8.17	38.64	14	182.6	−17.8	234.9	2.6	288.3	−87.3	1.4	2.7
SD-8	8.22	38.59	7	187.1	−11.8	192.5	4.4	325.8	−82.6	2.3	4.4
SD-9	8.31	38.55	9	354.5	−29.9	221.6	3.5	231.1	65.0	2.1	3.8
SD-10	8.3	38.55	10	173.3	−20.3	112.0	4.6	147.3	−83.0	2.5	4.8
SD-11	8.3	38.58	9	175.5	0.8	200.9	3.6	65.9	−80.2	1.8	3.6
SD-12	8.29	38.6	13	172.1	−20.7	147.0	3.4	146.3	−81.8	1.9	3.6
SD-13	8.28	38.62	14	168.4	2.8	78.0	4.5	89.1	−74.9	2.3	4.5
SD-14	8.25	38.64	14	185.1	−17.7	108.5	3.8	299.8	−84.9	2.1	4.0
SD-17	8.01	38.52	6	160.6	−34.6	110.2	6.4	160.9	−68.2	4.2	7.4
SD-19	7.94	38.37	14	173.9	−15.1	79.5	4.5	126.7	−84.0	2.4	4.6
SD-20	8.01	38.37	5	177.3	8.3	8.3	14.3	51	−77.5	7.2	14.4
SD-21	7.86	38.42	6	157.9	−15.4	115.3	6.3	129.8	−68.1	3.3	6.4
SD-24	7.93	38.27	10	162.0	−22.9	89.7	5.1	142.6	−71.8	2.9	5.4
SD-25	8	38.32	9	174.9	−8.1	106.1	5.0	89.2	−83.5	2.5	5.1
All mean except SD-4			19	355.7	9.9	22.9	7.3	273.3	84.6	40	5.4
Expected Mean direction (2Ma)			32	1.9	15.5	105.6	2.5	–	–	–	–
Dipole-BC03	–		32	–	–	–	–	182	86	96.6	3

Table columns headings: Site ID, Site name; long and lat, location coordinates in longitude and latitude; N, number of samples; Ds and Is, declination and inclination in stratigraphic coordinates; K, Precision parameter; α95, 95% confidence interval. SD = Silti-DebreZeyt. VGP = virtual geomagnetic pole. Since all the lava flows, we investigated are horizontal, there is no difference in the site mean directions in geographic and stratigraphic coordinates.

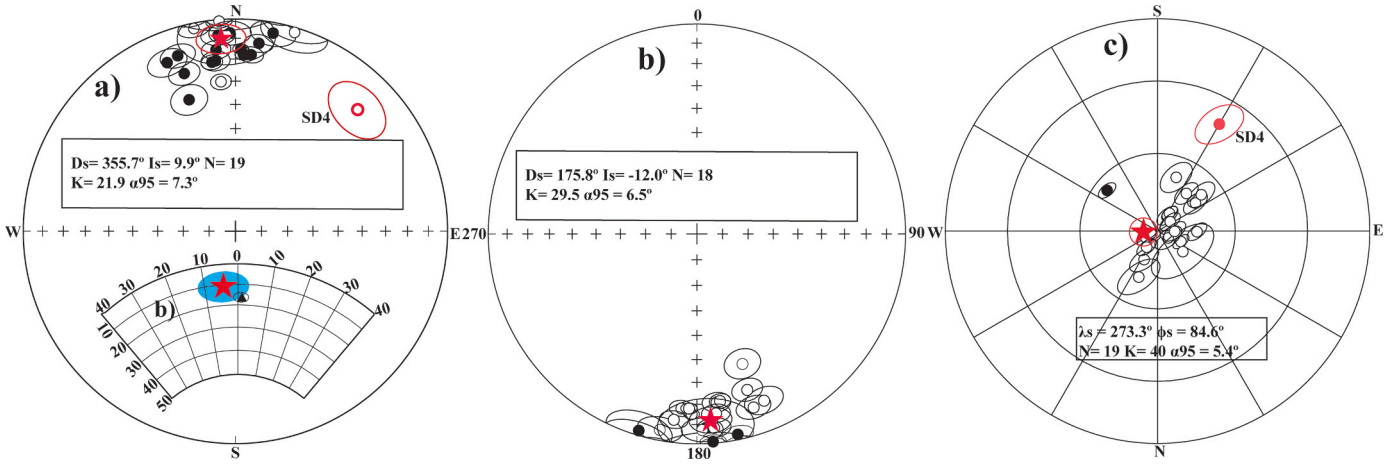


Fig. 6. Stereographic projection of site-mean directions for 19 sites in (a) and their corresponding overall site mean directions given by red star symbols; the red star with blue envelope in the bottom panel is the obtained mean direction with alpha95 envelope with their respective expected directions indicated with black triangle (Besse and Courtillot, 2003). (b) The overall sites mean directions for eighteen sites of reverse polarity represented by red star symbols with alpha95 (c) Stereographic projection of individual Virtual Geomagnetic Pole (VGP) indicated by full black circles and the overall mean VGP direction is shown in red star with corresponding 95% confidence circle around it.

of equivalent age in Butajira is considered, as discussed in Muhabaw et al. (2022). In contrast, Muhabaw et al. (2022) found that the Gedemsa magmatic segment in the Central Main Ethiopian Rift Valley underwent a 17° clockwise rotation during the Early Pleistocene, which is influenced by local stress fields and deformation interactions associated with the YTVL, Debre Zeyt volcanic lineament, and Wonji fault belt. Although the YTVL GBL lineaments are situated to the north and south of the present study area (Figs. 1 and 7), the area seems far from the strain interaction zones of the lineaments, which the paleomagnetic data also vindicates.

In the Main Ethiopian Rift (MER), fault patterns vary along its strike; in the Butajira volcanic zone, rift axis faults trend N10°E-N15°E while the Miocene Guraghe border faults trend N30°E-N40°E (Agostini et al., 2011b), and undergo an orthogonal and oblique extension respectively relative to the orientation of the plate divergence (095°–100°) (Agostini et al., 2011a; Keir et al., 2006). Besides, sinistral oblique shear evidenced by small-scale pull-apart, and echelon tensional fissures interconnected by short oblique-slip faults aligned with the MER border faults were reported (WoldeGabriel et al., 1990; Boccaletti et al., 1998), which could somehow trigger local block rotation. The observed counterclockwise rotation of the Butajira volcanic zone is likely due to the shift in extension direction from NW/SE, orthogonal to the rift shoulder, to ESE/WSW, oblique to the rift shoulder as the deformation migrate from the rift border striking NE/SW to the rift floor striking NNE/SSW.

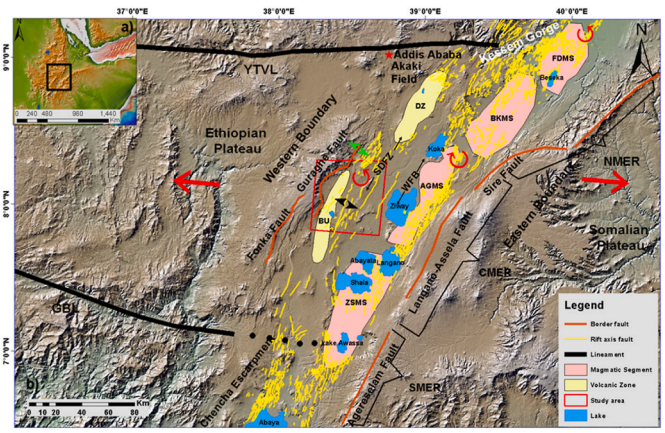


Fig. 7. The same DEM map with kinematic interpretation and sense of rotation in the study area and regions.

(Bonini et al., 1997, 2005; Boccaletti et al., 1998; Wolfenden et al., 2004; Casey et al., 2006).

5. Conclusion

The main magnetization carrier minerals for Early Pleistocene Butajira volcanic rocks are Ti-poor titano-magnetite, magnetite, titano-hematite, and minor Ti-poor titano-hematite, which carry identical directions, which prove their reliability on recording original magnetization direction. The TH and AF demagnetization techniques revealed two components of magnetization directions: the low stability (LS) and the high stability (HS) components. The LS component, representing a recent overprint magnetization, was eliminated through low-temperature (<330 °C) and field (<20 mT) demagnetization, while the HS component, obtained through higher temperature (>330 °C) and AF (20–80 mT) steps, is recognized to be the ChRM direction acquired during emplacement of the lava. The site's mean ChRM directions exhibit excellent directional data sets, as expected from recent volcanic materials (Table .1). on the paleomagnetic overall mean direction of the Butajira volcanic zone, D = 355.7°, I = 9.9° (N = 19, K = 22.9, α_{95} = 7.3°), and the expected mean geomagnetic diploe reference field directions for the African Plate, D = 1.9°, I = 15.5° (N = 32, K = 105.5, α_{95} = 2.5°) (Besse and Courtillot, 2003), are statistically different both at 99% and 95% significance level (McElhinny, 1964; McFadden and McElhinny, 1990), which results in declination anomalies, ΔD = 6.2° ± 6.2°, and Inclination anomaly, ΔI = 5.6° ± 6.2°. The inclination anomaly is a consistent result throughout the MER and Afar related to the effect of the long-standing non-dipole field (e.g., Kidane et al., 1999; Kidane et al., 2003). The calculated declination anomaly suggests a counterclockwise rotation of ~6°, which is in good agreement with the result of (Kidane et al., 2009, 2010; Nugsse et al., 2018) of the central MER about a vertical axis.

The MER is the best example of an oblique rift, where the rift is extending oblique to the trend of the rift (e.g., Corti et al., 2013). In the CMER, where the Butajira volcanic zone is located, fault patterns show a persistent N10°E-N15°E trend while the border faults (Miocene Guraghe border faults) trend N30°E-N40°E, with the rift extending in a 095°-100°direction (Agostini et al., 2011b; Keir et al., 2006). These data and observation confirm the rift axis faults, exhibiting pure-dip slip movement, open orthogonal to and the border faults oblique to the general Nubia-Africa plate divergence, with the latter resulting in local deflection of extension (Corti et al., 2013). Based on multidisciplinary data from field and seismic, analogue modeling done suggested minor clockwise rotation was necessary to accommodate the difference in slip-direction (Corti et al., 2013).

Hence, the observed CCW rotation of the Butajira volcanic zone is likely to have been caused by changes in extension direction and the deformation movement at the rift border. The two major lineaments north of and south of the study area did not have a significant effect on the vertical axis block rotation of the Butajira volcanic zone.

CRediT authorship contribution statement

Netsanet Mulugeta: Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tesfaye Kidane:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Kahsay Nugsse:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Geramu Fufa:** Writing – original draft, Resources, Project administration, Investigation, Funding acquisition. **Demise Tadessa:** Writing – original draft, Resources, Project administration, Investigation, Funding acquisition. **Ameha A. Muluneh:** Methodology, Software.

Declaration of competing interest

As authors of the paper titled "Paleomagnetic Evidence of Early Pleistocene Counterclockwise Rotation in the Butajira Volcanic Zone, Central Main Ethiopian Rift" we affirm that there are no financial or non-financial conflicts of interest that might have impacted the conception, conduct, or interpretation of the research presented in this paper.

Data availability

Data will be made available on request.

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